



MILESTONE 2 · DATASET PREPARATION

AI-Powered Driver Wellness & Safety Monitoring System

Five driver-monitoring modules — identified, cleaned, split without leakage, and structured model-ready for Milestone 3. Presented one member at a time.

Kushagra · Shiwani · Shubham · Sohini · Ravina

Five Modules, Five Owners

	 Shiwani Landmark Temporal Features	YawDD	EAR / MAR & head-pose sequences → LSTM
	 Kushagra Video-Based Fatigue Detection	UTA-RLDD	Safe / Caution / High-Risk over time
	 Shubham Driver Activity Classification	AUC Distracted Driver	5-class distraction CNN
	 Sohini Seat Belt & Phone Detection	DMS Dataset	2-class YOLO object detection
	 Ravina Smoking & Drinking Detection	YOLO smoking/drinking set	2-class YOLO object detection



YawDD Dataset

Yawning and driver-face videos feed extraction of facial landmarks, eye, mouth and head-pose signals — the behavioural signals of fatigue over time.

MODEL

LSTM (temporal sequence)

INPUT SHAPE

(30, 5) temporal feature sequence

KEY POINTS

- Extracts EAR, MAR, head pitch, yaw and roll per frame
- Signals represent behavioural change across time, not single frames
- Final feature format: 30-frame temporal window, 5 features/frame
- Feeds a temporal LSTM model in Milestone 3

Landmark Extraction with MediaPipe

99.35%

landmark extraction success rate across the full corpus

288,174

frames processed across 349 YawDD videos

286,289

valid frames retained after cleaning

1,885

invalid frames discarded (no landmark)

QUALITY CHECKS

- ✓ Head-pose stability validation
- ✓ EAR & MAR range validation
- ✓ Landmark jitter review
- ✓ Manual inspection of success & failure frames

PREPROCESSING

- Invalid frames removed before feature building
- Normalized with statistics from the training split only
- Fixed-length temporal windows of 30 frames, stride 15
- Output reliable for temporal sequence modeling

Subject-Wise, Leakage-Free

LEAKAGE PREVENTION

- All videos from one subject assigned to a single split
- Every 30-frame window stays inside its native video's split
- Z-score normalization computed only on training data
- Same stats then applied to validation & test

PROCESSED STRUCTURE

```
processed_dataset/  
├ raw_features_all_videos.csv  
├ normalized_features_all_videos.csv  
├ train_windows.npy · val · test  
└ train_meta.csv · val · test
```

Split · Subject-wise

Format · .npy + .csv

Hosting · Google Drive ✓



Kushagra

UTA-RLDD Dataset

Real-life drowsiness videos (Fold1 part1 & part2) — 12 subjects, 36 videos. Fatigue is a temporal condition, so behaviour is learned over time, not from single frames.

MODEL

CNN-LSTM · CNN-GRU · TCN · 3D CNN

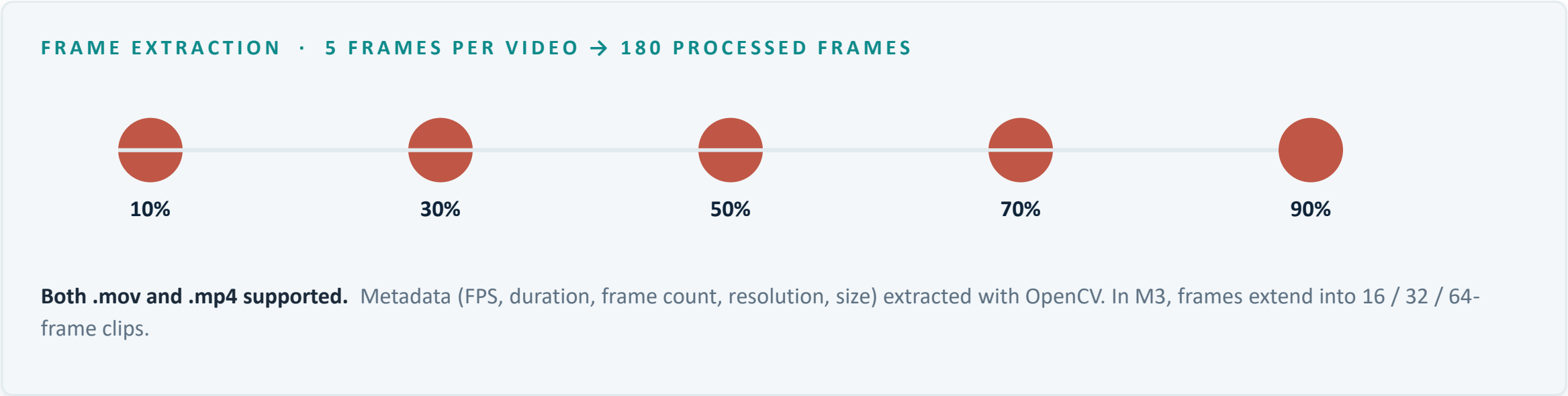
INPUT SHAPE

(16, 224, 224, 3) video sequence

KEY POINTS

- Original labels remapped: 0 → Safe, 5 → Caution, 10 → High Risk
- 12 Safe · 12 Caution · 12 High-Risk videos — perfectly balanced
- A single frame can't distinguish tired vs blinking vs talking
- Feeds CNN-LSTM / CNN-GRU / TCN / 3D-CNN in Milestone 3

Verified Clean, Frame-Sampled



Strict Subject-Level Split

Split	Subjects	Videos	Frames
Train	01 – 08	24	120
Validation	09 – 10	6	30
Test	11 – 12	6	30

All three videos of a subject stay in the same split — frames are extracted only after the split is finalized, so no frame from one video ever crosses splits.

CLASS-WISE FOLDERS

```
fatigue_detection/  
├─ train/  
│   ├─ Safe/  
│   ├─ Caution/  
│   └─ High_Risk/  
├─ val/    (same 3)  
└─ test/   (same 3)
```

Split · Subject-wise

Format · Frame folders

Hosting · Google Drive ✓



Shubham

AUC Distracted Driver

Image-based classification of what the driver is doing — a CNN activity classifier over five real-world driving activity classes.

MODEL

ResNet50 · MobileNetV3 · EfficientNet-B0

INPUT SHAPE

(3, 224, 224) RGB image

KEY POINTS

- Classes: safe_driving, texting_phone, talking_phone, turning, other_activities
- Supports ResNet50, MobileNetV3 and EfficientNet-B0
- 4,307 clean images retained after quality cleaning
- Resized to 224×224 and augmented for CNN training

Cleaning & Augmentation

2,969

duplicate images

Removed · perceptual hashing

10

corrupted images

Removed · OpenCV load check

31

blurry images

Retained · realistic conditions

4,307

clean images

Final retained dataset

PREPROCESSING

- Resize all images to 224×224
- Normalization applied during model training
- Blur scored with Laplacian variance

AUGMENTATIONS

- Horizontal flip
- Random brightness & contrast
- Random rotation

70 / 15 / 15 by Class

TRAIN / VALIDATION / TEST

Train 70%

Validation 15%

Test 15%

LEAKAGE PREVENTION

- Duplicate images removed before splitting
- Class-wise splitting keeps class balance
- Post-split hash check flagged near-duplicate images
- Recommend sequence-/subject-wise split once driver IDs exist

CLASS-NAMED FOLDERS

```
processed_dataset/  
├─ train/  {5 class folders}  
├─ validation/  {5 class folders}  
└─ test/  {5 class folders}
```

Link is updated.



DMS Dataset

A filtered Driver Monitoring System set turned into a clean two-class YOLO detector for visible safety-compliance objects in the cabin.

MODEL

YOLOv8n · YOLO11n

INPUT SHAPE

(640, 640, 3) RGB image + YOLO labels

KEY POINTS

- Two classes — 0: Phone, 1: Seatbelt
- Irrelevant classes removed; remaining classes remapped
- 6,140 images with 6,140 matching label files
- 6,701 bounding boxes validated for correctness

Image-Label Parity & Validation

6,140

images, each with a matching label file

6,701

bounding boxes validated

2

clean target classes: Phone, Seatbelt

VALIDATION CHECKS

- ✓ Every image has a corresponding annotation file
- ✓ YOLO bounding boxes follow the expected format
- ✓ Normalized coordinates stay within valid bounds
- ✓ Irrelevant classes removed; kept classes remapped
- ✓ Unannotated background frames filtered out

PREPROCESSING

- Images resized to 640 × 640
- Annotations validated against YOLO schema
- Classes remapped to a clean 2-class layout
- Compliance-relevant objects only retained

70 / 20 / 10, Sequence-Aware

TRAIN / VALIDATION / TEST

Train 70%

Val 20%

Test 10%

LEAKAGE PREVENTION

- Sequence-based stratified split
- Original split boundaries preserved
- No random frame-level migration between splits
- Prevents memorizing scene-specific backgrounds

YOLO STRUCTURE

```
dms_processed_yolo/  
├ data.yaml  
├ train/  images/  · labels/  
├ valid/  images/  · labels/  
└ test/   images/  · labels/
```

Hosting · Google Drive ✓



YOLOv8 Detection Pipeline

Two independently-labelled YOLO datasets merged into one training-ready detector. Classes are harmonized by name — {smoking → 0, drinking → 1} — dropping every non-target class.

MODEL

YOLOv8-based detector

INPUT SHAPE

(640, 640, 3) RGB image + YOLO labels

KEY POINTS

- 18,932 source images across two datasets (A + B)
- Name-based mapping prevents mislabelled boxes leaking in
- 7,159 non-target boxes (hands-on-wheel etc.) dropped
- Run on Albumentations 2.0.8 · OpenCV 4.13.0

Clean, but Sharply Imbalanced

0 / 0 / 0

corrupt · malformed · unmapped records

15,900

clean target instances across 12,909 images

66.6%

of boxes cover < 1% of the frame (small-object heavy)

1.23

objects per image on average — sparse scenes

CLASS BALANCE · 19.6× IMBALANCE



KEY EDA FINDINGS

- 15,127 smoking vs 773 drinking instances
- Small objects dominate → recommend imgsz 832–960
- Objects cluster middle-upper frame (driver-facing camera)
- Drinking is single-source (dataset B) — domain-shift risk

Leakage-Safe, Training-Ready



STRATIFIED 80 / 10 / 10 SPLIT

Split	Images	Smoking	Drinking
Train	9,910	11,696	636
Val	1,239	1,447	64
Test	1,239	1,441	73

Deduplicated (521 removed) before the split — val & test stay augmentation-free.

AUGMENTATION · TRAIN ONLY

Bbox-aware Albumentations, 2 base variants/image. Minority class drinking gets +17 augmented copies per image to counter the 19.6× imbalance.

EXPORT

12,388 originals + 24,538 augmented

Ultralytics-ready · data.yaml · 3,060 MB archive · Google Drive ✓